DevOrbis_Final - Requirement Analysis Report

Document Type: QA / SQA Requirement Analysis
Project: DevOrbis_Final - AI Operations Hub
Application URL: https://dev-orbis-final-frontend.vercel.app/
Source Baseline: DevOrbis_Final Software Requirements Specification (SRS)
Version: Draft v1.0
Analysis Role: SQA / QA Requirement Analysis
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Status: Draft for stakeholder clarification

1. Executive Summary

This report analyzes the DevOrbis_Final - AI Operations Hub from an SQA perspective to determine whether its requirements are clear, complete, consistent, unambiguous, and testable enough to support test design and execution.

The review is based on the DevOrbis_Final SRS, which specifies authentication, authorization, chat, RAG, lead management, task management, AI-assisted task creation, dashboards, integrations, and deployment.

The SRS is a strong baseline. It defines roles, workflows, AI/RAG architecture, data model, and technology stack. However, several requirements need clarification before QA can build a fully reliable test basis.

Overall QA Assessment

Area	Assessment
Requirement coverage	Broad - authentication, roles, chat, RAG, leads, tasks, AI task creation, dashboards, integrations all covered
Testability	Moderate - many requirements testable; some lack precise expected behaviour
Consistency	Needs verification - some cross-module behaviours not fully specified
Completeness	Needs clarification - field-level and UI-level details incomplete
QA readiness	Partially ready - usable for draft test design; critical ambiguities must be resolved before sign-off

Requirement Quality Criteria Used

Clarity, Completeness, Consistency, Testability, Traceability.

2. Scope Reviewed

Area	Coverage
Public site	Landing page, Sign Up, Login
Area	**Coverage**
User role	Home, AI Text Chat, AI Voice Chat, Logout
Staff role	Staff Dashboard, My Tasks, Create Task
Admin role	Admin Dashboard, Knowledge Base, All Tasks, Task Detail, Edit Task, AI Usage, AI Conversations, Create Task
Cross-cutting	Sidebar navigation, role-based navigation, Logout
Functional modules	Authentication, Authorization, Chat, RAG, Leads, Tasks, AIAssisted Task Creation, Dashboards, AI Usage, Knowledge Base, Integrations, Voice AI

3. Detailed Functional Requirement Analysis

Status values: **Clarity** (Yes / Partial / No), **Testable** (Yes / Partial / No), **Complete** (Yes / Partial / No).

3.1 Authentication & Account Management

Req ID	Requirement Summary	Clarity	Testable	Complete
FR-AUTH-01	User registration with username, email, password	Yes	Yes	Partial
FR-AUTH-02	Email verification after registration	Yes	Yes	Partial
FR-AUTH-03	Login with email and password	Yes	Yes	Yes
FR-AUTH-04	JWT-based authentication on login	Yes	Yes	Yes
FR-AUTH-05	JWT stored in HTTPonly cookie `agentic_rag_access_token`	Yes	Yes	Yes
FR-AUTH-06	JWT contains user ID and role	Yes	Yes	Yes
FR-AUTH-07	User logout	Yes	Yes	Partial

FR-AUTH-08	Admin and Staff accounts may be manually marked verified	Yes	Yes	Partial
FR-AUTH-09	Normal users expected to verify email	Yes	Yes	Partial
FR-AUTH-10	Backend authentication middleware	Yes	Yes	Yes

Observations: Password rules, username/email validation, verification token expiry, and logout redirect not specified.

3.2 Authorization & Role Management

Req ID	Requirement Summary	Clarity	Testable	Complete
FR-AUTHZ-01	Role-based authorization middleware	Yes	Yes	Yes
FR-AUTHZ-02	Roles: admin, staff, user	Yes	Yes	Yes
FR-AUTHZ-03	Backend enforces access control; frontend reflects it	Yes	Yes	Partial
FR-AUTHZ-04	Staff limited to permitted functionality	Yes	Yes	Partial
FR-AUTHZ-05	Admin has highest access	Yes	Yes	Partial

Observations: Exact permission matrix and cross-role access behaviour not specified.

3.3 User Management

Req ID	Requirement Summary	Clarity	Testable	Complete
FR-USER-01	User registers an account	Yes	Yes	Partial
FR-USER-02	User authenticates	Yes	Yes	Yes
FR-USER-03	User interacts with chatbot	Yes	Yes	Yes
FR-USER-04	User maintains conversations	Yes	Yes	Yes

FR-USER-05	User views own relevant task info where applicable	Yes	Yes	Partial
FR-USER-06	User may become lead after AI qualification	Yes	Yes	Partial
FR-USER-07	Maximum one active lead per user	Yes	Yes	Yes

3.4 Chat & Conversations

Req ID	Requirement Summary	Clarity	Testable	Complete
FR-CHAT-01	Conversational AI chatbot	Yes	Yes	Yes
FR-CHAT-02	Create new conversations	Yes	Yes	Yes
Req ID	**Requirement Summary**	**Clarity**	**Testable**	**Complete**
FR-CHAT-03	Continue existing conversations	Yes	Yes	Yes
FR-CHAT-04	Socket.IO for real-time communication	Yes	Yes	Partial
FR-CHAT-05	Chat history persisted in database	Yes	Yes	Yes
FR-CHAT-06	Manual chat history (no LangGraph checkpointer)	Yes	Partial	Partial
FR-CHAT-07	Message sender types: Human, AI	Yes	Yes	Yes
FR-CHAT-08	Chat title from first few words of first message	Yes	Yes	Partial
FR-CHAT-09	Timezone-aware timestamps	Yes	Yes	Partial

Observations: Socket event payloads, title generation rule, and retention policy not specified. **3.5 AI Assistant**

Req ID	Requirement Summary	Clarity	Testable	Complete

FR-AI-01	Assists with conversational interaction	Yes	Partial	Partial
FR-AI-02	Assists with knowledge retrieval	Yes	Yes	Yes
FR-AI-03	Assists with task generation	Yes	Yes	Yes
FR-AI-04	Assists with lead qualification	Yes	Partial	Partial
FR-AI-05	Assists with routing/ workflow decisions	Yes	Partial	Partial
FR-AI-06	Not presented as fully autonomous	Yes	N/A	Yes
FR-AI-07	AI output does not bypass normal workflow	Yes	Yes	Yes

3.6 Knowledge Base

Req ID	Requirement Summary	Clarity	Testable	Complete
FR-KB-01	Admin manages knowledge base	Yes	Yes	Yes
Req ID	**Requirement Summary**	**Clarity**	**Testable**	**Complete**
FR-KB-02	Documents processed for AI retrieval	Yes	Yes	Partial
FR-KB-03	Administrative source of organizational information	Yes	Yes	Yes
FR-KB-04	Formats, chunking, embeddings follow implementation	Yes	Partial	Partial

3.7 RAG & Retrieval

Req ID	Requirement Summary	Clarity	Testable	Complete
FR-RAG-01	RAG to answer from organization knowledge base	Yes	Yes	Yes

FR-RAG-02	Retrieval and reranking in pipeline	Yes	Yes	Partial
FR-RAG-03	Reranker reorders chunks before generation	Yes	Partial	Partial
FR-RAG-04	Responses grounded in retrieved information	Yes	Yes	Partial
FR-RAG-05	pgvector for vector search	Yes	Yes	Yes
FR-RAG-06	Gemini for embeddings and LLM	Yes	Yes	Yes

3.8 Lead Management

Req ID	Requirement Summary	Clarity	Testable	Complete
FR-LEAD-01	Qualified users represented as leads	Yes	Yes	Yes
FR-LEAD-02	Every lead corresponds to a user	Yes	Yes	Yes
FR-LEAD-03	At most one active lead per user	Yes	Yes	Yes
FR-LEAD-04	Lead created only after AI qualification	Yes	Partial	Partial
FR-LEAD-05	Priority values: High, Medium, Low	Yes	Yes	Yes
FR-LEAD-06	Lead includes priority, status, summary, user association	Yes	Yes	Partial
Req ID	**Requirement Summary**	**Clarity**	**Testable**	**Complete**
FR-LEAD-07	Status values based on implementation	Partial	Partial	Partial

3.9 AI Lead Qualification

Req ID	Requirement Summary	Clarity	Testable	Complete
FR-LEADQ-01	AI participates in lead qualification	Yes	Partial	Partial

FR-LEADQ-02	Flow: Interaction -> AI conversation -> Qualification -> Lead -> Follow-up	Yes	Partial	Partial
FR-LEADQ-03	Exact algorithm follows implementation	Partial	Partial	Partial

3.10 Task Management

Req ID	Requirement Summary	Clarity	Testable	Complete
FR-TASK-01	Tasks created for leads	Yes	Yes	Yes
FR-TASK-02	Tasks assigned to staff	Yes	Yes	Yes
FR-TASK-03	Status values: pending, in_progress, completed	Yes	Yes	Yes
FR-TASK-04	Priority values: low, medium, high	Yes	Yes	Yes
FR-TASK-05	Task fields: title, description, priority, status, lead, staff, timestamps	Yes	Yes	Yes
FR-TASK-06	New tasks start pending	Yes	Yes	Yes
FR-TASK-07	Backend validates task creation	Yes	Yes	Partial

3.11 AI-Assisted Task Creation

Req ID	Requirement Summary	Clarity	Testable	Complete
FR-AITASK-01	Natural language task description	Yes	Yes	Yes
FR-AITASK-02	AI generates title, description, priority	Yes	Yes	Yes
FR-AITASK-03	Generated info placed in normal task form	Yes	Yes	Yes
Req ID	**Requirement Summary**	**Clarity**	**Testable**	**Complete**
FR-AITASK-04	AI does not bypass normal workflow	Yes	Yes	Yes

FR-AITASK-05	Final creation through standard flow and validation	Yes	Yes	Yes

3.12 Dashboards & Analytics

Req ID	Requirement Summary	Clarity	Testable	Complete
FR-DASH-01	Admin dashboard operational overview	Yes	Yes	Partial
FR-DASH-02	Caching for expensive aggregate stats	Yes	Yes	Yes
FR-DASH-03	Staff dashboard assigned task status overview	Yes	Yes	Partial

3.13 AI Usage

Req ID	Requirement Summary	Clarity	Testable	Complete
FR-AIUSE-01	Admin AI Usage area	Yes	Yes	Yes
FR-AIUSE-02	Operational visibility into AI usage	Yes	Yes	Partial
FR-AIUSE-03	Exact metrics follow implementation	Partial	Partial	Partial

3.14 n8n Automation

Req ID	Requirement Summary	Clarity	Testable	Complete
FR-N8N-01	Backend communicates with n8n via webhooks	Yes	Yes	Partial
FR-N8N-02	Backend triggers n8n for task create/update/ delete	Yes	Yes	Partial
FR-N8N-03	n8n handles Monday.com integration	Yes	Yes	Partial

3.15 Monday.com Integration

Req ID	Requirement Summary	Clarity	Testable	Complete
FR-MON-01	Task events trigger n8n	Yes	Yes	Partial

FR-MON-02	n8n handles external Monday.com integration	Yes	Yes	Partial
FR-MON-03	Backend does not own entire sync logic	Yes	Yes	Yes
FR-MON-04	No full Monday -> app sync at this stage	Yes	Yes	Partial

3.16 Voice AI

Req ID	Requirement Summary	Clarity	Testable	Complete
FR-VOICE-01	AI Voice Chat feature	Partial	Partial	Partial

Note: The application includes an AI Voice Chat page. Functional behaviour, supported browsers, STT/TTS configuration, and expected outcomes require clarification before detailed test design.

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4. Detailed Non-Functional Requirement Analysis

4.1 Performance

Req ID	Requirement Summary	Clarity	Testable	Complete
NFR-PERF-01	Responsive CRUD interaction	Partial	Partial	No
NFR-PERF-02	Responsive chat interaction	Partial	Partial	No
NFR-PERF-03	Numerical latency targets not assumed	Yes	Yes	Yes

4.2 Scalability

Req ID	Requirement Summary	Clarity	Testable	Complete
NFR-SCALE-01 to 06	Architecture supports future expansion of agents, users, leads, tasks, integrations, KB content	Yes	Partial	Partial

4.3 Reliability

Req ID	Requirement Summary	Clarity	Testable	Complete

NFR-REL-01	Handle API failures gracefully	Yes	Yes	Partial
NFR-REL-02	Handle AI failures gracefully	Yes	Yes	Partial
NFR-REL-03	Handle external integration failures gracefully	Yes	Yes	Partial
NFR-REL-04	Handle database errors gracefully	Yes	Yes	Partial

4.4 Security

Req ID	Requirement Summary	Clarity	Testable	Complete
NFR-SEC-01	Authentication for protected resources	Yes	Yes	Yes
NFR-SEC-02	Role-based authorization	Yes	Yes	Yes
NFR-SEC-03	HTTP-only cookies	Yes	Yes	Yes
NFR-SEC-04	Backend authorization (not frontend-only)	Yes	Yes	Yes
NFR-SEC-05	Input validation (Zod + Yup)	Yes	Yes	Partial
NFR-SEC-06	Protect against unauthorized data access	Yes	Yes	Partial
NFR-SEC-07	Secure handling of AIgenerated content	Yes	Partial	Partial
NFR-SEC-08	Protect external integration credentials	Yes	Partial	Partial
NFR-SEC-09	Protect environment secrets	Yes	Partial	Partial
NFR-SEC-10	No fabricated compliance certifications	Yes	N/A	Yes

4.5 AI Security

Req ID	Requirement Summary	Clarity	Testable	Complete

NFR-AISEC-01	Prompt injection protection	Yes	Partial	Partial

Req ID	Requirement Summary	Clarity	Testable	Complete
NFR-AISEC-02	Malicious retrieved document protection	Yes	Partial	Partial
NFR-AISEC-03	AI workflow manipulation protection	Yes	Partial	Partial
NFR-AISEC-04	Unauthorized access through conversation protection	Yes	Partial	Partial
NFR-AISEC-05	Data leakage protection	Yes	Partial	Partial
NFR-AISEC-06	Authorization independent of AI	Yes	Yes	Yes
NFR-AISEC-07	AI output does not replace backend authorization	Yes	Yes	Yes

4.6 Maintainability

Req ID	Requirement Summary	Clarity	Testable	Complete
NFR-MNT-01	Modular codebase (frontend/backend/AI/ integration)	Yes	Partial	Partial
NFR-MNT-02	Specified technology stack	Yes	Yes	Yes
NFR-MNT-03	Extensible code	Yes	Partial	Partial

4.7 Extensibility

Req ID	Requirement Summary	Clarity	Testable	Complete
NFR-EXT-01 to 05	Future support for multi-agent, voice, integrations, advanced RAG, additional modules	Yes	Partial	Partial

4.8 Data Privacy

Req ID	Requirement Summary	Clarity	Testable	Complete
NFR-PRIV-01 to 08	Appropriate handling of user, auth, conversation, lead, task, KB, AI-generated content; no fabricated compliance	Yes	Partial	Partial

4.9 Auditability / Traceability

Req ID	Requirement Summary	Clarity	Testable	Complete
NFR-AUD-01	Important actions traceable via app data	Yes	Partial	Partial
NFR-AUD-02	No fabricated audit system	Yes	N/A	Yes
NFR-AUD-03	Audit logging identified as future improvement	Yes	N/A	Yes

4.10 Usability

Req ID	Requirement Summary	Clarity	Testable	Complete
NFR-USA-01	Loading states	Yes	Yes	Partial
NFR-USA-02	Error states	Yes	Yes	Partial
NFR-USA-03	Empty states	Yes	Yes	Partial
NFR-USA-04	Toast-based validation feedback	Yes	Yes	Partial
NFR-USA-05	Clear role-specific navigation	Yes	Yes	Partial
NFR-USA-06	Responsive UI (desktop + smaller screens)	Yes	Yes	Partial
NFR-USA-07	Accessible interactive controls	Yes	Partial	Partial

4.11 Error Handling

Req ID	Requirement Summary	Clarity	Testable	Complete

NFR-ERR-01	Centralized backend error handling	Yes	Yes	Yes
NFR-ERR-02	Frontend errors via UI states and toasts	Yes	Yes	Partial
NFR-ERR-03	Meaningful errors; no sensitive internals	Yes	Yes	Partial

4.12 Caching

Req ID	Requirement Summary	Clarity	Testable	Complete
NFR-CACHE-01	Redis for caching only, not primary storage	Yes	Yes	Yes
Req ID	**Requirement Summary**	**Clarity**	**Testable**	**Complete**
NFR-CACHE-02	Admin dashboard statistics cached	Yes	Yes	Yes
NFR-CACHE-03	Cache key `admin:dashboard:stats`	Yes	Yes	Yes
NFR-CACHE-04	TTL 60 seconds	Yes	Yes	Yes
NFR-CACHE-05	Cache invalidated after successful aggregate mutations	Yes	Yes	Partial
NFR-CACHE-06	Local dev uses `redis://localhost:6379`	Yes	Yes	Yes
NFR-CACHE-07	Production uses Redis Cloud	Yes	Yes	Partial
NFR-CACHE-08	Redis accessed via `REDIS_URL`	Yes	Yes	Yes

4.13 Real-time (Socket.IO)

Req ID	Requirement Summary	Clarity	Testable	Complete
NFR-RT-01	Socket.IO for real-time chat	Yes	Yes	Yes
NFR-RT-02	Socket authenticated via cookie/JWT	Yes	Yes	Yes

NFR-RT-03	Authenticated user stored in socket session	Yes	Partial	Partial
NFR-RT-04	Events: chat:message, chat:response, chat:error	Yes	Partial	Partial

4.14 Forms & Validation

Req ID	Requirement Summary	Clarity	Testable	Complete
NFR-FORM-01	React Hook Form + Yup	Yes	Yes	Yes
NFR-FORM-02	Zod for backend validation	Yes	Yes	Yes
NFR-FORM-03	Migrated forms: signup, login, create task, AI task	Yes	Yes	Yes
NFR-FORM-04	Toast-based validation errors	Yes	Yes	Partial
Req ID	**Requirement Summary**	**Clarity**	**Testable**	**Complete**
NFR-FORM-05	Create-task uses shared RHF instance	Yes	Yes	Yes
NFR-FORM-06	AI task component updates form via setValue	Yes	Yes	Yes

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5. Business Rules Analysis

5.1 Lead Domain

Rule ID	Business Rule	Clarity	Testable	Consistent
BR-LEAD-01	Every lead corresponds to a user	Yes	Yes	Yes
BR-LEAD-02	Not every user is a lead	Yes	Yes	Yes

BR-LEAD-03	Max one active lead per user	Yes	Yes	Yes
BR-LEAD-04	Lead created only after AI qualification	Partial	Partial	Partial
BR-LEAD-05	Priority: High / Medium / Low	Yes	Yes	Yes
BR-LEAD-06	Lead fields: priority, status, summary, user	Yes	Yes	Partial
BR-LEAD-07	Status values from implementation	Partial	Partial	Partial

5.2 AI Lead Qualification

Rule ID	Business Rule	Clarity	Testable	Consistent
BR-LEADQ-01	AI participates in qualification	Yes	Partial	Partial
BR-LEADQ-02	Qualification based on conversation + business criteria	Partial	Partial	Partial
BR-LEADQ-03	Exact algorithm not invented	Yes	N/A	Yes

5.3 Task Domain

Rule ID	Business Rule	Clarity	Testable	Consistent
BR-TASK-01	Tasks created for leads	Yes	Yes	Yes
Rule ID	**Business Rule**	**Clarity**	**Testable**	**Consistent**
BR-TASK-02	Tasks assignable to staff	Yes	Yes	Yes
BR-TASK-03	New tasks start `pending`	Yes	Yes	Yes
BR-TASK-04	Status: pending / in_progress / completed	Yes	Yes	Yes
BR-TASK-05	Priority: low / medium / high	Yes	Yes	Yes
BR-TASK-06	Admin can create and assign	Yes	Yes	Yes
BR-TASK-07	Staff can create per permissions	Partial	Partial	Partial
BR-TASK-08	AI does not bypass workflow	Yes	Yes	Yes

BR-TASK-09	Final creation through backend validation	Yes	Yes	Yes

5.4 Chat & AI

Rule ID	Business Rule	Clarity	Testable	Consistent
BR-CHAT-01	Chat history persisted in DB	Yes	Yes	Yes
BR-CHAT-02	Manual chat history (no LangGraph checkpointer)	Yes	Partial	Partial
BR-CHAT-03	Sender types: Human / AI only	Yes	Yes	Yes
BR-CHAT-04	No separate role column	Yes	Yes	Yes
BR-CHAT-05	Title from first few words	Partial	Partial	Partial
BR-CHAT-06	Timezone-aware timestamps	Yes	Yes	Partial

5.5 RAG

Rule ID	Business Rule	Clarity	Testable	Consistent
BR-RAG-01	KB questions answered from retrieved information	Yes	Yes	Partial
BR-RAG-02	Reranking applied to improve retrieval	Yes	Partial	Partial
Rule ID	**Business Rule**	**Clarity**	**Testable**	**Consistent**
BR-RAG-03	Responses grounded in retrieved context	Yes	Yes	Partial
BR-RAG-04	Context evaluated and reordered before generation	Yes	Partial	Partial

5.6 Authentication & Authorization

Rule ID	Business Rule	Clarity	Testable	Consistent
BR-AUTH-01	Protected resources require auth	Yes	Yes	Yes

BR-AUTH-02	Role-based access enforced on backend	Yes	Yes	Yes
BR-AUTH-03	Frontend reflects access restrictions	Yes	Yes	Partial
BR-AUTH-04	Normal users expected to verify email	Yes	Yes	Partial
BR-AUTH-05	Admin / Staff manually verified	Yes	Yes	Partial
BR-AUTH-06	JWT contains user ID and role	Yes	Yes	Yes
BR-AUTH-07	Cookie name `agentic_rag_access_token`	Yes	Yes	Yes
BR-AUTH-08	Cookie is HTTP-only	Yes	Yes	Yes

5.7 Knowledge Base

Rule ID	Business Rule	Clarity	Testable	Consistent
BR-KB-01	Only Admin manages knowledge base	Yes	Yes	Yes
BR-KB-02	Documents processed for AI retrieval	Yes	Yes	Partial
BR-KB-03	Formats/chunking/ ingestion from implementation	Partial	Partial	Partial

5.8 Integrations

Rule ID	Business Rule	Clarity	Testable	Consistent
BR-INT-01	Task events trigger n8n	Yes	Yes	Partial
BR-INT-02	n8n handles Monday.com integration	Yes	Yes	Partial

Rule ID	Business Rule	Clarity	Testable	Consistent
BR-INT-03	Backend does not own full sync	Yes	Yes	Yes
BR-INT-04	No full Monday -> app sync	Yes	Yes	Partial
BR-INT-05	App is primary source of task state	Yes	Yes	Yes

5.9 Role-Based Navigation

Rule ID	Business Rule	Clarity	Testable	Consistent
BR-NAV-01	Admin nav items	Yes	Yes	Partial
BR-NAV-02	Staff nav items	Yes	Yes	Partial
BR-NAV-03	User nav items	Yes	Yes	Partial
BR-NAV-04	Backend + frontend enforce access	Yes	Yes	Partial

5.10 Caching

Rule ID	Business Rule	Clarity	Testable	Consistent
BR-CACHE-01	Redis only for caching	Yes	Yes	Yes
BR-CACHE-02	Invalidate on mutation	Yes	Yes	Partial
BR-CACHE-03	TTL 60 seconds	Yes	Yes	Yes

5.11 AI Task Generation

Rule ID	Business Rule	Clarity	Testable	Consistent
BR-AITASK-01	AI generates title/ description/priority	Yes	Yes	Yes
BR-AITASK-02	Populates normal form, no bypass	Yes	Yes	Yes
BR-AITASK-03	User reviews/edits before submission	Yes	Yes	Yes
BR-AITASK-04	Final creation through backend validation	Yes	Yes	Yes

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6. Recommended Acceptance-Criteria Additions

The following additions are recommended to make requirements fully testable.

6.1 Authentication

Invalid email format -> defined error, no account created Weak password -> minimum rules defined and enforced Duplicate email / username -> defined error, no duplicate created

- Invalid credentials -> generic error, no email disclosure
- Unverified account login -> define blocked/allowed behaviour
- Logout -> cookie cleared, redirect to defined page
- Session expiry -> duration and behaviour defined
- Verification token -> expiry and resend behaviour defined

6.2 Authorization

- Cross-role URL access -> defined response (403 / redirect / not found)
- Unauthenticated access -> redirect to login
- Missing/invalid JWT -> 401
- Wrong role -> 403
- Role assignment and manual verification flows defined

6.3 Chat

- New chat title generation rule (exact)
- Previous chats sorted by last updated
- History loaded on chat open
- Socket failure -> defined error and retry
- Socket event schema defined (payloads)
- Empty message blocked
- AI failure -> defined message and retry

6.4 AI Assistant

- Timeout duration and fallback message
- Service unavailable handling
- Out-of-scope query handling
- Ambiguous query handling
- Where AI output is reviewed before operational action

6.5 Knowledge Base

- Supported file types (PDF, DOCX, TXT)
- Max file size
- Duplicate upload behaviour
- Corrupted file behaviour

- Delete confirmation and embedding cleanup
- List columns defined

6.6 RAG

- Top-K definition
- No-relevant-doc behaviour defined • Reranker applied and reorders
  Grounding verified per claim
  Citation includes document + section/page
  Fallback response for empty retrieval
- Decline rather than fabricate

6.7 Lead Management

- Qualification trigger conditions
- Priority mapping from qualification
- Exact status enum values
- Database-level uniqueness enforcement
- Admin and Staff lead list behaviour
- Summary source and format

6.8 Task Management

- Required fields and validation
- Status transition rules
- Reassignment rules
- Delete rules (soft/hard, who)
- Title/description length limits
- List filters and pagination

6.9 AI-Assisted Task Creation

- Free-text input accepted
- AI generates all three fields
- Fallback for empty/invalid AI output
- Editable before submission
- Standard backend validation after edit
- AI failure -> error and retry

6.10 Dashboards

- Admin metrics defined
- Task Overview counts defined
- Cache TTL and invalidation behaviour
- Staff metrics defined
- Refresh behaviour after mutations
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6.11 AI Usage

- Tokens displayed: total, input, output
- Cost calculation defined
- Refresh interval defined

6.12 Integrations

- Webhook trigger events defined •

Payload schema defined

Retry policy and fallback defined One-way sync confirmed

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- Conflict behaviour if Monday record edited

6.13 Voice AI

- Supported browsers and permissions
- Recording behaviour and storage
- STT accuracy expectations
- TTS voice/pacing expectations
- Tools available during voice interaction
- Expected business outcomes (task, lead, etc.)
- Call ID scheme and recording access

6.14 Non-Functional

- Performance latency targets (TBD)
- HTTPS enforcement
- Cookie flags (HTTP-only + Secure)
- Rate limiting on public endpoints
- Prompt injection detection and rejection
- Redis / AI / DB / n8n failure fallbacks
- Loading, error, and empty states per page
- Responsive breakpoints
- Accessibility level

Total recommended additions: ~119.

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7. QA Traceability Approach

7.1 Traceability Chain

```
Requirement ID -> Test Scenario ID -> Test Case ID -> Test Execution Result -> Defect ID
```

7.2 ID Conventions

Level	Format	Example
Requirement	FR--NNN / NFR--NNN	FR-CHAT-001
Test Scenario	SCN-NNN	SCN-001
Test Case	TC-NNN	TC-001
Voice Test Case	VOICE-NNN	VOICE-001
Execution	TC-NNN-RUN	TC-001-RUN1
Defect	BUG-NNN	BUG-001

7.3 Coverage Metrics

Metric	Definition
Requirement Coverage	Requirements with >= 1 scenario + 1 test case / Total
Scenario Coverage	Scenarios with >= 1 test case / Total
Test Execution Coverage	Executed / Planned
Pass Rate	Passed / Executed
Defect Density	Defects per module
Defect Closure Rate	Resolved / Logged
Stage-Level Attribution	Failures by pipeline stage

7.4 Defect Traceability Fields

Defect ID, Linked Requirement ID, Linked Scenario ID, Linked Test Case ID, Module, Failure Stage (ingestion / chunking / retrieval / context / generation / tool selection / tool execution / post-processing), Severity, Priority, Status, Evidence.

7.5 Bidirectional Traceability

- Forward: Requirement -> Scenario -> Test Case -> Execution -> Defect
- Reverse: Defect -> Test Case -> Scenario -> Requirement

Orphan defects and uncovered requirements are both signals of gaps.

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8. QA Readiness & Sign-Off Recommendation

8.1 Readiness by Module

Module	Readiness
Authentication	Ready (password rules TBD)
Authorization	Partially Ready (permission matrix TBD)
User Management	Ready
Chat	Ready (event schema TBD)
AI Assistant	Partially Ready (error handling TBD)
Knowledge Base	Partially Ready (formats/ingestion TBD)
RAG	Partially Ready (reranking, grounding, citation TBD)
Lead Management	Partially Ready (status enum, qualification trigger TBD)
AI Lead Qualification	Partially Ready (algorithm, thresholds TBD)
Task Management	Ready (transitions TBD)
AI-Assisted Task Creation	Ready
Dashboards	Partially Ready (metrics TBD)

AI Usage	Partially Ready (metric definitions TBD)
Module	**Readiness**
Integrations	Partially Ready (webhook contract TBD)
Role-Based Navigation	Ready
Voice AI	Partially Ready (behaviour TBD)

8.2 Critical Clarifications Required

Blockers:

- Lead status enum values
- AI qualification trigger conditions
- Task status transition rules
- Cross-role URL access behaviour
- Email verification mandatory vs optional
- Password policy
- Knowledge Base supported file types and max size
- RAG behaviour when no relevant chunks retrieved
- Voice AI functional behaviour
- Socket event payload schema

High Priority: Session expiry, unverified login response, dashboard metrics, AI Usage metric definitions, webhook payload schema, reranking algorithm, citation format, task reassignment rules, Staff task-creation permissions, KB deletion propagation.

Medium Priority: Chat title rule, timezone display, cache invalidation on partial failure, Redis fallback, AI generation timeout, toast wording, per-page states, responsive breakpoints, accessibility level, logout redirect.

8.3 Recommended Document State

- v1.0 - Current draft
- v1.1 - Critical clarifications resolved
- v1.2 - High priority clarifications resolved
- Final - Approved by stakeholder, ready for test design sign-off

8.4 Sign-Off Recommendation

The SRS is suitable as a working baseline for initial test design but **should not be treated as the final unambiguous test oracle** yet. Critical and High priority clarifications must be resolved and versioned. Medium priority items should be tracked in a clarification register. UI-observed features not covered in the SRS must be explicitly included in scope or marked out of scope.

Once complete, QA can proceed with confidence in test scenarios, test cases, execution, defect reporting, and final test summary.

8.5 Final Statement

The DevOrbis_Final - AI Operations Hub is a well-architected AI-assisted platform with a strong functional specification. With the clarifications above addressed, the project is ready for systematic QA evaluation across authentication, authorization, role-based workflows, chat, RAG, lead management, task management, AI-assisted task creation, dashboards, AI Usage, integrations, and Voice AI.